BFU10: Database Development

Assignment 3 of 3

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SECTION 1: DESCRIBE THE MAIN REASONS WHY ORGANISATIONS USE DATABASES TO:

Improve Productivity:

Databases are used in organizations to improve productivity as they are much more efficient than physical records, data that the user needs can be accessed with a few simple clicks and the results are returned almost instantly, which can easily be turned into a report for printing, which compared to physical records; gathering specific data could take a very long time, especially if it's a large specific set of data they need, and, if they want to make a report they would need to create it manually, massively hindering their productivity.

Make Decisions:

Organizations use databases to help make decisions as they provide clean, reliable, and instant data access, which can easily be turned into physical reports to display the data specified, which can then be easily sent over to management. The fast and easy access allows for close time-critical decisions to be made correctly according to the data, which in most cases in large organizations, this would be incredibly difficult or impossible without a database and only physical records.

Present Information:

Because databases are incredibly effective and efficient in helping present information, they can quickly gather specific data within seconds which can be put into a report to present information effectively, which is all done within a few minutes, reducing wasted time, money, and resources.

Interpret Data:

Because with databases you can specify in queries what specific data you are after, for example you could query for “what users joined this year has their first name start with B” or “which patients haven't had an appointment for over a year” which is very useful for organizations as they can easily interpret the data they need, instead of having to ignore a record that doesn't meet the criteria, which also loops back to improving productivity.

Perform Calculations:

Because they take out the manual labour of manually calculating record by record, and the desired calculations can be completed and outputted within a few seconds, which the manual labour in large datasets could take hours, and mistakes can occur which could set that progress back or end with an incorrect calculation, especially if it's a very small group of people.

Manage Large Datasets:

Because large datasets in physical copies are very difficult to manage, even with filing cabinets that separate data it could still be hard to find specific data you are looking for, and the cabinets will take up lots of space which could take up multiple rooms, wasting space that could be used for something more useful. Databases can be contained into just one computer, or be stored elsewhere on the cloud; meaning no unnecessary space is taken up from the datasets.

SECTION 2: EXPLAIN THE PURPOSE OF THE FORTH COFFEE SUPPLIES DATABASE AND HOW IT IS USED BY THE ORGANISATION

The purpose of the database is to provide Forth Coffee Supplies an easy-to-use digital system to store/manage records of their customers, employees, orders, etc. It is likely used to help manage their data for their customers, as it compared to physical records it is easy to grab all of a customer's orders within a few clicks, which can then be turned into a summary report which would could be useful for sharing orders that need processing to a delivery team, sharing with management, customer service, etc.

The organization uses the database with their customers, employees, and orders data all filled in order to effectively provide service to their customers, they seem to use it to manage orders with their customers, such as checking if payment is received, when was the order was shipped, when did they pay, all of the pricing statistics, their information, which all of this is integrated to help the company work efficiently.

SECTION 3: EXPLAIN THE PURPOSE OF THE DIFFERENT TOOLS AND TECHNIQUES AND HOW THEY ARE USED IN THE FOURTH COFFEE SUPPLIES DATABASE AND HOW THEY HELP IMPROVE PRODUCTIVITY, ACCURACY AND USABILITY.

Navigation:

The purpose of navigation in Access is to make it easy to use and reduce its complexity for its users, navigation can be implemented in many ways throughout Access, by default Access has built in navigation buttons in forms at the bottom to help navigate through records (if there are any), FCS use the built-in navigation buttons and their own throughout the database, the manage orders form has navigation buttons to run queries, forms, reports, and macros. The built-in navigation buttons are used in some pages (such as manage orders) to move back and forth through records easily. These buttons improve productivity as they are easily within reach wherever the user is on the form, and don't require you to look at the cramped overwhelming menus in Access at all, which also improves the usability for its users, especially for those who aren't familiar with Access. The navigation also has its hand in improving accuracy as they give confidence the user is in the right place for where they need to be.

Table Structures used to split the data into logical tables:

The purpose of table structures is for storing data regarding a subject, such as all of Fourth Coffee Supplies customers data gets stored in their Customers table, and the customers order data is stored in the Orders table, etc. The data then can be easily split using queries and forms returning specifically what the user wants, we can see this used throughout FCS's database with their tables being used in queries to manage data easily, improving productivity and usability as they cleanly separate datasets for easy use, making it less confusing, which also in turns helps accuracy because the separation of the data in the tables make it much more organized and simple, reducing mistakes such as entering users confidential notes in a random customers order.

Field Characteristics:

The purpose of the field characteristics is to define specific settings for a field, which those settings control what data type it is, its size limit, the format that's expected to be inputted, default value, validation rules and texts, if its required, etc. These are used in the Fourth Coffee Supplies database as they are essential in the usage and creation of a database, an example of the usage of field characteristics usage is inside the customers table at the ZIPCode field, you can see they have set the field size to have a limit of 50 characters, and have an input mask of "00000\-9999" which they are used to reduce the possibility of incorrect data being entered such as something invalid such as "95431585950381" which would not fit in the field data. Field characteristics help improve productivity because the field characteristics help guide the user into the right direction for entering valid data, they greatly improve accuracy because of its powerful validation rules and input masks blocking incorrect inputs, and they make the database more usable, as the field characteristics make the database more user friendly and accessible to new users.

Validation Rules:

The purpose of validation rules is to ensure invalid data cannot be entered, such as a rule that only accepts an input as long as it contains an "@", one place where I saw a validation rule used in the Fourth Coffee supplies database was for discount in order details containing the rule "< 1" which would only allow numbers less than 1 to be inputted, which improves accuracy as it makes it rejects invalid data.

Indexing

Indexing provides a unique identification value for every record, which prevents exact copies of records being added into the database, without indexing duplicates could be created which leads to incorrect data as operations may fail to act on the right one, and it can be confusing for a user to find a customer they're looking for if somebody else has the same name, which indexing solves with its unique value on every record. Which all of this helps improve productivity (less time spent finding right data), accuracy (makes it easy to find duplicates, and the right data), and usability, as they reduce the complexity of managing the data. Fourth Coffee supplies use it in their database to uniquely identify their customers, which in the customers table all customers have a "CustomerID" which is a number that increments with every new customer, which makes them easy to identify.

Records:

The purpose of records is to store data in a table, they are a single row in the table that contain a specific dataset, for example in Customers of Fourth Coffee Supplies database, there is a record of a customer and their relevant information, such as their Customer ID, First Name, Last Name, Phone Number, etc. They are there to cleanly separate data into their own relevant use case to prevent mismatched data, in this case, they contain customer information. These help improve productivity, accuracy and usability as they make it easy to understand where data is meant to be, no time is wasted finding a customer's phone number, if its on the same record you'll know that's the data for them, which all links back to making it usable.

Relationships:

The purpose of relationships in a database are to integrate table data together, making it so indexed records can be easily referenced for later usage, this is utilized in the Fourth Coffee Supplies database in many places, one example is that they have a relationship built between CustomerID in Customers, and CustomerID in Orders, they use this as it links together records in orders to their respected customers, making it easy to view data from both sides and ensures that the customer is valid, helping accuracy. Relationships also help improve productivity and usability as they simplify connections between tables, making complex queries, confusing searches, irrelevant.

Forms:

The purpose of forms are to create a user interface for the database, which they can be used to make it less overwhelming for a new user as they are typically used as an overlay for managing the database, which the designs can be tailored for any user, which this helps productivity as they can reduce the amount of actions the user needs to take for a task, they can help accuracy and usability as you can integrate in the forms design labels which contain information to help guide a user on how to use and manage it correctly. Forth Coffee Supplies use forms in their database to manage their customers, orders, products, etc, easily. For example, in the form “Manage Orders” they have dropdowns of their customers which when there is a customer selected it shows all of their order history in a table, disclosing all relevant information for easy access and management, on the side they also integrated macros to help simplify the form even more.

Queries:

Queries purpose in the database are to grab specific pieces of data regarding the users specifications, for example, “Customers in last 6 months that have placed orders” which the query would parse the data with the requirements and return the customers that meet the criteria containing their CustomerID, First Name, Last Name, Phone Number, and the sum of orders that they have done. Forth Coffee Supplies use queries to return all sorts of data, an example query of theirs I will describe is “Order Details Totals” which uses the “Order Details Extended” query to group by OrderID, and creates a sum of the Line-Item Totals, which when ran displays the total of each order. This improves productivity as it helps condense data down to what the user needs, instead of manually combing through data to find what they’re after. It also helps accuracy as you can only see what you need, which also links back into improving the usability.

Reports:

Reports purpose is outputting sets of data onto a small page in a neat, easy to understand format, which they can be easily shared as a file or printed out to be shared to those that want the data, removing the hurdles of database permissions, manually inputting into a spreadsheet/word document, etc. Reports make it simple as it creates that “spreadsheet” of data for you ready to be printed or sent. This improves productivity as less time is wasted manually creating the report, and staff no longer have to do the monotonous task of manually creating the report, which overtime could take longer and longer to do, reducing their productivity. They also improve accuracy as reports can easily be given out to other staff which could verify themselves manually if the data is correct. Usability is massively improved with reports as they can be handed out to anyone, as they don’t need access to the database or training on how to use it. FCS use reports in their database to list their customers and all of the orders in a summary, which they likely use these reports to share data with other staff in the organization that don’t have access to the database but for example want an order summary to figure out what task needs to be completed.

Macros:

Macro’s purpose is to automate/execute tasks within the databases user interface, macros can automatically run if a value changes, a value equals something, etc, the usual use case for macros is for opening a specific form upon clicking a button. FCS use macros in their database to make the buttons in their manage orders form have functionality, for example clicking the “Add or Delete Customers” opens up the “Add or Delete Customer” form for the user. Macro’s improve productivity as they simplify tasks for the user by reducing the number of clicks, or doing it all automatically for them. They also help accuracy as they can automatically update content for the user without having to be done manually, usability is also improved because macros can be utilized so the user never has to touch anything outside of the forms and reports, avoiding the complicated Access interface entirely.

SECTION 5: LOOK AT THE FOURTH COFFEE SUPPLIES DATABASE IN MORE DETAIL AND DISCUSS ITS STRENGTHS AND WEAKNESSES

Strengths:

The database has the strengths of having all of the forms and reports easily within reach, the dropdowns at the top of the manage orders form give quick access to data, the dropdowns also display the available options which makes it easy to find what you are after. On the right all of the forms and reports are easily accessible within a button click. The design also has a neat professional setting within most of its forms, the reports have a neat layout that is also easy to understand and cleanly lays out the data.

Weaknesses:

Some forms are completely missing the designs and are just bare tables, it is not always clear on how to navigate through some forms, the manager orders form navigation bar at the bottom can be easily missed and can be confusing, the form to manage customers also uses this navigation bar and the form does not contain a button to create a new customer so you have to manually click the button in the navigation bar, which some might not know about. Upon opening the database, you are landed directly into the manage orders form, which I feel the database lacks a main menu as being dropped straight into manage orders was confusing until I learnt that it is an all-in-one form, the “Manage Orders” banner makes the form appear as if its meant to be a separate page dedicated only to orders, this creates confusion and should be changed.

The database also has a major flaw, it is almost completely missing validation rules throughout all the tables, some even without input masks, this allows invalid data to be entered easily, massively reducing data integrity and could land the organization in legal troubles and or send wrong orders.

SECTION 6: EXPLAIN THE PURPOSE OF EVANS MANUFACTURING DATABASE AND HOW IT IS USED BY THE ORGANISATION

The purpose of the Evans Manufacturing Database is to digitally manage and store data regarding training of their employees, which consists of the employee's contact details, their qualifications, and their role in the company. They also store records of their Event Suppliers, training events they do, and training events that have took place. This type of use-case is best suited to be used with a digital database as it would be difficult to manage this type of data in a paper-based system and would be more difficult to send peoples records and less cost effective, which I believe EM chose to use this digital database system because of that. EM uses the database to record their employees training via logging events they attended, which they can then store as proof it has been completed in case needed later and to show they meet legal requirements. They can easily figure out who needs to attend a training event as there's a button that opens a report displaying employees that have yet to attend one with their details.

SECTION 7: EXPLAIN THE PURPOSE OF THE DIFFERENT TOOLS AND TECHNIQUES AND HOW THEY ARE USED IN THE EVANS MANUFACTURING DATABASE AND HOW THEY HELP IMPROVE PRODUCTIVITY, ACCURACY AND USABILITY.

Table Structures:

The purpose of table structures is to cleanly separate and organize data for ease of use and cleanliness. In the EM database they are used to separate data that is more effectively stored elsewhere, such as an employees attended training events are not stored in "tblEmployee" but in "tblRecord" which allows for easy management of multiple records of training events, which this also helps accuracy as trying to store multiple records in one field will cause problems with usability and integrity later on. The table structures improve productivity as the separation of content makes it easier to find what you are looking for.

Field Characteristics:

Field characteristics serve the purpose of defining what data in a field can be, such as only being a number, Boolean, short text, or a validation rule for example being ">0" only allowing numbers greater than 0. EM's database uses field characteristics to define specific criteria for the data in the field, or to help guide the user into what it's for, an example being StartDate in employees which is set to Date/Time which onlys allows a date or time to be entered, not allowing other inputs. This improves productivity, accuracy and usability because it makes it more understandable for the average user, as it makes it impossible to input invalid data with the settings that can be set, and the terms used to describe the characteristics are also simple to understand without being overwhelming.

Validation Rules:

The purpose of validation rules is to ensure invalid data cannot be entered, such as a rule that only accepts an input as long as it contains an “@”, in EM’s database they used validation rules to restrict invalid data entry, for example in the employess table the first name field has a validation rule like “(Not Like **[!a-z]**) And (Len([FirstName])>=3 And Len([FirstName])<=30)” which ensures only letters can be entered and must have 3 to 30 characters. This improves productivity as it reduces errors and going back to fix them overtime and makes it more user friendly, improving usability and accuracy.

Indexing

The purpose of Indexing is creating a unique identifier for a row of data, making it easier and more accurate regarding finding data as you don’t have to rely on information that might have duplicates as it is not unique. Which all helps improve productivity as less times wasted going through a data set finding what you need, without indexing it could be easy to make mistakes and would be harder to find data. An example of indexing in EM’s database is in the employees table, which they have indexing on the EmployeeID to give every employee a unique identifier that goes up incrementally.

Records:

The purpose of records is to store a specific data set within one row, such as in a customer’s table there will be a customer record that only contains data related to them, this helps accuracy and usability as the clean separation of data makes sure you don’t get confused on what your trying to get, and helps prevent data leaks, also in turn improving productivity. In EM’s database an example of them using records in the the employees table which you can see a record of an employee's data contained in a single row, containing their ID, first name, last name, address, etc.

Relationships:

The purpose of relationships in a database is to help integrate tables together, this simplifies tasks and the tables as an example, in an orders table you would find OrderID, and then CustomerID, you can create a relationship with that CustomerID with the customers table, which integrates that customers and orders together for easy use in queries and for helping validity, improving accuracy, usability, and productivity. An example of relationships used in EM’s database is that they have linked together EmployeeID with tblRecords, giving seamless integration for seeing what training events that employee has attended.

Forms:

The purpose of forms is to provide a customizable user interface to interact with the database, which can be tailored for any use, they can be tailored for those with accessibility needs, organizations that need minimal training, and or for efficiency. Which can all help productivity and usability, forms can also help with accuracy as they can be made to easily guide the user on how to use them, reducing data accuracy problems. EM's database has forms for navigation also, for example the whole main menu is a form, which contains buttons that can take you to sub forms for managing employees, event suppliers, training events, etc.

Queries:

Queries purpose in the database are to grab specific pieces of data regarding the users specifications, for example, "Customers in last 6 months that have placed orders" which the query would parse the data with the requirements and return the customers that meet the criteria containing their CustomerID, First Name, Last Name, Phone Number, and the sum of orders that they have done. Queries improve productivity as they help condense data down to what the user needs, instead of manually combing through data to find what they're after. It also helps accuracy as you can only see what you need, which also links back into improving the usability. EM's database uses queriesfor example to gather data on employees of theirs that have not attended a training event, which we see used in the report.

Reports:

Reports purpose is to simplify data and prepare into a sheet, which can easily be shared or printed into as a physical copy, this is so people that don't have access to the database can still be given data that they need, or to help the user understand what they're looking at. Which the ease and simple function improve usability for all, and productivity as it takes out the hard work of gathering the data yourself and helps accuracy because it won't make mistakes gathering the data manually unlike a person which could make mistakes if its monotonous. EM uses reports in their database to display printable lists of courses their employees have been on, and the ones that havent been on a course yet.

Macros:

Macro's purpose is to automate/execute tasks within the databases, they can be configured to automatically run a task on changes of data or with a user clicking a button, the macro can execute multiple tasks for them. This massively helps productivity as it reduces the number of unnecessary clicks and simplifies its usability, which can also help with accuracy. EM uses macros in their database within their forms, for example the create new employee button is a macro that takes you to the end of the records so you can input into a clean empty record.

Navigation

Navigations purpose in the database is to help change through records and into other menus easily, with no need for complicated workarounds to just open a form, they are just a simple click to move around such as clicking a next button to change a customer, which this improves productivity from the simplicity given, and its usability, which will in turn improve accuracy. Navigation examples can be easily found in EM's database in their forms; they have buttons to help navigate through employees easily.

SECTION 9: LOOK AT THE EVANS MANUFACTURING DATABASE IN MORE DETAIL AND DISCUSS ITS STRENGTHS AND WEAKNESSES

Strengths:

It is easy to multitask, and is seemingly encouraged as there is a button on the main menu that automatically opens all forms and minimizes them for you for easy access and multitasking, the main menu also is helpful for displaying forms and reports you can go to compared to FCS. There is also an easy access button to record an employee attending a training event which is useful as it reduces unnecessary clicks. On data entry forms there are also notices that tell you requirements of data entry in a form which is helpful in understanding the database.

Weaknesses:

The colour scheme is atrocious, not just by preference, it creates many usability and accessibility issues and looks very unprofessional for a manufacturing company, text is often black on blue backgrounds which makes it very hard to read, even for those with good eyesight. The mix of extremely bright colours and poor content contrast can cause eye strain very quickly. The layout is also all over the place, the layout is shifted out of place and looks like it was just quickly put together, it's also quite cramped. The forms while some have notices to help you, some of them are not done correctly, as for example, County and Country can easily have invalid data entered, and prior qualifications are also just typed out instead of being put into a table which can easily be problematic, and the roles dropdown for employees is very confusing you are given an option to input a role ID that has no reference, or use the very small list. The search function for finding courses and employee is on brings up an input box asking for the employees first name, which blocks you from pressing behind it so you can't quickly find the name without cancelling it. Searching by name can also be unreliable as somebody could also have the same name, yet there is no option to input an employee ID.